

BROWNVILLE AGCM, AB, Canada

WMO: 712830

Lat: 56.1178N

Lon: 117.8864W

Elev: 748

StdP: 92.66

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.1	-29.1	-35.6	0.1	-32.0	-32.6	0.2	-29.0	11.6	0.8	10.2	-1.1	1.7	20	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	27.2	15.9	25.6	15.1	24.0	14.6	17.5	24.2	16.6	23.3	15.7	22.0	3.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	11.7	20.1	14.0	10.9	19.1	13.1	10.3	18.1	52.2	24.5	49.4	23.2	46.6	22.0	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.5	7.4	-36.1	30.0	2.9	1.2	-38.2	30.8	-39.9	31.5	-41.5	32.2	-43.6	33.1		
(5)			-36.3	19.4	2.9	1.3	-38.3	20.4	-40.0	21.2	-41.6	21.9	-43.7	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-11.5	-11.5	-6.1	2.9	10.5	14.3	16.2	15.1	10.2	3.1	-6.9	-12.9
	DBStd	12.36	8.80	8.34	7.80	5.00	4.38	3.13	2.59	3.26	4.41	4.90	7.07	7.98
	HDD10.0	3527	668	603	498	218	47	4	1	3	50	218	507	711
	HDD18.3	5975	926	837	756	463	243	125	76	107	244	471	757	969
	CDD10.0	615	0	0	0	5	63	133	192	161	56	5	0	0
	CDD18.3	21	0	0	0	0	1	4	9	7	1	0	0	0
	CDH23.3	479	0	0	0	3	42	91	172	135	36	0	0	0
	CDH26.7	72	0	0	0	0	7	15	21	27	3	0	0	0
	WSAvg	3.2	3.4	3.1	3.4	3.6	3.7	3.2	2.9	2.9	3.0	3.2	2.9	3.1
	PrecAvg	413	24	16	19	19	41	74	70	48	36	22	23	21
(15) Precipitation	PrecMax	540	56	35	52	47	91	135	129	125	104	61	54	71
	PrecMin	251	5	4	2	1	16	19	27	18	0	4	5	2
	PrecStd	91	14	8	11	11	18	35	26	27	21	13	15	15
	PrecStd	91	14	8	11	11	18	35	26	27	21	13	15	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.6	10.3	22.1	27.3	28.8	28.7	29.6	26.6	20.7	8.8	5.3
		MCWB	2.4	2.5	4.8	10.2	13.0	16.3	16.9	16.6	15.2	11.7	4.9	2.1
	2%	DB	4.9	4.2	8.0	16.8	24.0	25.6	26.9	26.7	24.1	15.4	5.8	3.3
		MCWB	2.2	1.5	3.5	7.3	11.7	14.7	16.4	16.1	14.2	8.9	2.9	0.7
	5%	DB	3.6	2.5	5.7	13.2	21.4	23.9	25.3	24.5	21.0	12.8	4.3	1.0
		MCWB	1.4	0.3	2.0	5.6	10.4	13.9	15.6	15.4	12.6	7.0	1.7	-1.0
	10%	DB	1.6	0.3	4.0	10.7	19.5	22.0	23.7	22.7	18.3	10.5	2.3	-1.4
		MCWB	-0.2	-1.5	0.8	4.5	9.7	13.3	15.2	14.8	11.5	5.7	0.1	-3.0
	0.4%	WB	3.0	2.8	5.4	11.1	14.1	17.7	19.6	18.4	16.5	12.3	5.5	2.3
		MCDB	5.4	5.2	9.6	22.0	23.8	26.1	26.6	25.2	24.6	19.1	8.2	4.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.2	1.6	3.7	8.1	12.6	16.3	17.7	17.3	14.8	9.6	3.3	0.6
		MCDB	4.5	3.7	7.8	16.3	20.9	23.5	23.6	24.2	22.0	14.6	5.9	3.0
	5%	WB	1.4	0.3	2.2	6.0	11.4	15.0	16.8	16.4	13.4	7.5	1.7	-1.0
		MCDB	3.6	2.5	5.4	12.2	19.6	21.5	23.1	23.0	19.9	11.7	3.8	1.0
	10%	WB	-0.3	-1.5	1.0	4.7	10.4	13.9	15.9	15.4	12.1	6.2	0.4	-3.1
		MCDB	1.7	0.3	3.8	10.1	18.2	20.5	22.1	21.3	17.9	10.0	2.1	-1.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.4	10.5	13.0	11.8	12.5	12.7	12.4	9.1	8.0	8.4
		MCDBR	7.8	9.5	11.4	14.8	16.7	15.8	15.8	15.9	17.1	13.0	9.2	9.7
		MCWBR	6.1	7.5	7.8	7.8	7.5	7.0	7.0	6.9	8.1	7.6	6.9	8.1
		MCDBR	7.9	9.3	10.9	14.0	15.0	13.4	14.1	14.1	16.1	12.4	9.2	9.8
	5% WB	MCWBR	6.4	7.4	7.6	7.7	7.0	6.5	7.1	7.0	8.0	7.6	7.1	8.3
		MCWBR	6.4	7.4	7.6	7.7	7.0	6.5	7.1	7.0	8.0	7.6	7.1	8.3
(40) Clear-Sky Solar Irradiance	taub		0.245	0.242	0.287	0.318	0.333	0.343	0.344	0.341	0.303	0.277	0.236	0.220
	taud		2.235	2.351	2.310	2.392	2.419	2.410	2.423	2.418	2.536	2.561	2.417	2.223
	Ebn at Noon		681	858	885	905	909	900	891	870	867	812	717	624
	Edh at Noon		52	69	94	102	105	108	105	98	76	57	43	40
(44) All-Sky Solar Radiation	RadAvg		0.70	1.68	3.21	4.69	5.60	6.00	5.84	4.75	3.25	1.73	0.76	0.48
	RadStd		0.07	0.14	0.26	0.28	0.41	0.48	0.37	0.41	0.26	0.21	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57	N/A

Nomenclature: See separate page